Programming for Data Science

Data Science process with Python

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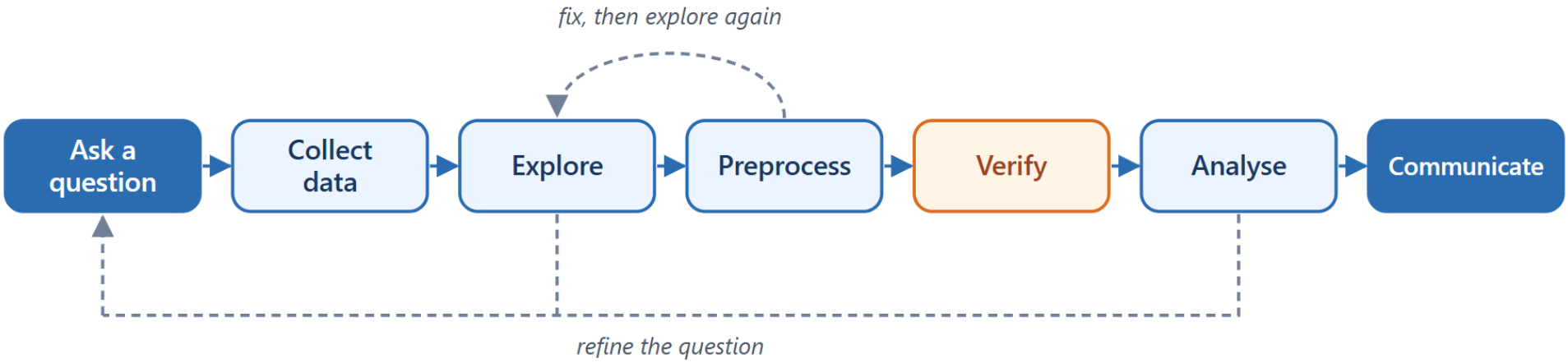
Introduction

- Today
  1. **The data science process**, in more detail: why we explore, preprocess and verify
  2. **Case study:** *How proficient is our class in Python?*
  3. **Types of data**: the type decides which statistics make sense
  4. **Descriptive statistics**: center, spread, shape
  5. **Demo:** the whole process in pure Python → `04-Python1Demo.ipynb`

Data science process

The data science process

- It's a **loop**, not a straight line
- Exploring often sends you **back**: to preprocess, to collect more data, or to change the question
- **Verify**: converting data without errors is not the same as converting it correctly



What is the purpose of exploring data?

To **understand the data**, so that we can:

1. **Find problems** in the data, then
    - Preprocess right away, so we can keep exploring, or
    - Go back and collect better data
2. **Refine the question**, or ask new questions this data *can* answer

- *You can't ask a good question about data you haven't looked at.*

Exploring tabular data: a checklist

Ask	Watch out for
How many rows and columns?	
What does one row mean?	Rows with a **different meaning** (totals, averages, headers repeated)
Are there duplicated rows?	The same record counted twice
What does each column mean?	Units, scales, abbreviations
What type does each column have now?	Numbers stored as **text**, dates as text
Numerical column: missing values? min? max?	**Impossible** values (negative age, score > max)
Categorical column: missing values? how many distinct values? a few examples?	Typos, inconsistent spelling (HCM, HCMC, ho chi minh)

Exploring tabular data

- Previous slide shows basic info about data we should explore
- In complex cases, we may want to explore additional info about data
- For example, if we want to know more about column distribution, we can compute additional info using descriptive statistics

Why preprocess the data?

- Two different reasons, at two different moments:

When	Goal	Example (quiz data)
During exploration	fix a problem so we *can keep exploring*	convert "8.29" (text) to 8.29 (number)
After choosing an analysis	prepare the data for *that specific method*	remove attempts that lasted days before comparing time vs. grade

- The second kind depends on the question, so it comes **after** refining the question.

Verify before you analyse

- Conversion can **succeed** and still be **wrong**. Check facts that *must* hold:
  - Totals add up (grade = sum of question scores)
  - Values agree with each other (duration = end time − start time)
  - Row counts make sense after every filter

Case study: the Python quiz

Example from previous lecture

- **Ask a question**: How is the Python proficiency in the class?
- **Collect data**: let students do quiz about Python in moodle, results can be downloaded as a csv file

```
State,Started on,Completed,Time taken,Grade/10.00,Q.
1 /0.53,…,Q. 19 /0.53
Finished,8/10/2025 11:29,8/10/2025 11:40,11 mins 56
secs,4.74,0.26,0.53,…
```

- One row per **attempt** (names and IDs removed)
- 19 questions, 0.53 points each, grade out of 10

What exploring revealed*

Finding	Consequence
70 rows, but only **66** are finished attempts	3 attempts were never submitted, and the last row is Moodle's **class average** → filter them out, explicitly
Everything is **text**	convert scores, dates and durations
- means "no value"	turn it into `None`, **not** 0
Some attempts took **hours or days**	the quiz was left open; that time isn't working time

- None of these would cause an error. All of them would change the results.

Refining the question

- *"How proficient is the class?"* is too vague to compute. Exploring tells us what the data **can** answer:

1. How are the **total grades** distributed?
2. Which **questions** were hardest (lowest average score)?
3. Is there a relationship between **time taken** and **grade**?
4. Are there **outliers**, students with unusual grades?

Files for today

- `04-Python1Demo.ipynb`: the notebook
- `Data/PythonQuiz_CQ_2026.csv`: the quiz results

- Put the CSV in a `Data/` folder next to the notebook.

Types of data

Question 1

1. How are the **total grades** distributed?

How do we answer question 1?

- Option 1: read all 66 grades and "get a feeling"
- Option 2: **summarise** them → **descriptive statistics**

Answer the question

- Different types of data will often have methods different descriptive statistics methods
- Common types of data:
  - Numerical data

    For example: scores, temperature
  - Categorical data
    - **Nominal**

      For example: color (*Blue, Red, Green*)
    - **Ordinal**

      For example: satisfaction level (*poor* < *fair* < *good* < *excellent*)

The type decides which statistics make sense

	Frequency, mode	Median, percentiles	Mean, standard deviation
Nominal	☑	☒ no order	☒
Ordinal	☑	☑	☑ only if the gaps are equal
Numerical	☑ (after grouping into bins)	☑	☑

Quick check: what type is it?

- Student `ID 22120001`
- Number of questions answered correctly
- Satisfaction rating `1–5` from a course survey
- Time taken: `11 mins 56 secs`
- Question number (`Q. 5`)

Descriptive statistics

What are Descriptive Statistics?

- **Descriptive statistics** are methods used to summarize and describe the main features of a dataset.
- They help you understand your data at a glance without making predictions or inferences about a larger population.
- Two main goals:
  - **Summarize** the data in a meaningful way
  - **Describe** patterns, trends, and characteristics

Three levels of summary

Level	Question	Categorical	Numerical
1. Center	what's a typical value?	mode	mean, median, (mode)
2. Center + spread	how much do values vary?	relative frequencies	mean & standard deviation; quartiles
3. Full distribution	what's the shape?	frequency table, **bar chart**	**histogram**, (box plot)

- Each level adds information, and each level can **hide** something the next one shows.

Categorical data: frequency & relative frequency

- *Favourite programming language* (survey, 125 students):

Language	Frequency	Relative frequency
Python	45	45 / 125 = **36%**
Java	30	24%
JavaScript	25	20%
C++	15	12%
Others	10	8%
Total	**125**	**100%**

- **Relative frequencies** let you compare groups of different sizes (a class of 40 vs. a class of 125).
- From our quiz: State → Finished: 66, In progress: 3, (empty): 1

Categorical data: mode

- **Mode** = the most frequent category. Above: **Python** (45).
  - **Unimodal**: one most frequent category
  - **Bimodal** / **multimodal**: two / several categories tie for most frequent
  - **No mode**: every category appears equally often
- For numerical data the mode is less useful: with continuous values (7.24, 7.37, …) almost every value is unique.

Level 1: Mean

- Give numerical data consisting of elements with values respectively: $v_1, v_2, \ldots, vn$
- $mean = \frac{1}{n}\sum_{i=1}^{n} v_i$

Level 1: median and percentiles

- The ***p*-th percentile** (0 <= p <= 100) is a value with about **p% of the data below it**.
  - The 75th percentile of the quiz grades is 9.6" → about 75% of students scored below 9.6
- **Median** = 50th percentile: half the values are below it, half above.

Computing a percentile

- There are different ways to compute ***p-th percentile***, here is one way:
- 1. **Sort** the values: $x_0 \leq x_1 \leq \cdots x_{n-1}$
- 2. Compute the position

$$k = \frac{p}{100}(n-1)$$

- 3. If ***k*** is a whole number $\rightarrow$ the answer is x_k. Otherwise $\rightarrow$ **interpolate** between $x_{\lfloor k \rfloor}$and $x_{\lceil k \rceil}$

Percentile examples

- **Odd count:** 1, 5, 3, 4, 2 → sorted 1, 2, 3, 4, 5
  - median: k = 0.5 * 4 = 2 → = **3**
- **Even count:** $1, 5, 3, 4, 2, 6$ → sorted $1, 2, 3, 4, 5, 6$
  - median: k = 0.5 * 5 = 2.5 → halfway between 3 and 4 → **3.5**
  - 75th percentile: k = 0.75 * 5 = 3.75 → 4 + 0.75 * (5 - 4) = **4.75**

Median vs mean

- Given this list: 1, 2, 3, 4, 5
  - Median = 3
  - Mean = 3
- Given this list: 1, 2, 3, 4, **500**
  - Median = 3
  - Mean = **102**
- Median is less affected by **outlier** (a value lying far away from the range of most values) than mean!

Meand & standard deviation

- The **standard deviation** is the average amount of ***variability*** in your dataset.
  - It tells you, on average, how far each value lies from the mean.
- A high standard deviation ➔ values are generally far from the mean
- A low standard deviation ➔ values are clustered close to the mean.
- Given numerical data consisting of elements with values respectively: $v_1, v_2, \ldots, vn$

$$SD = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(v_i - mean)^2}$$

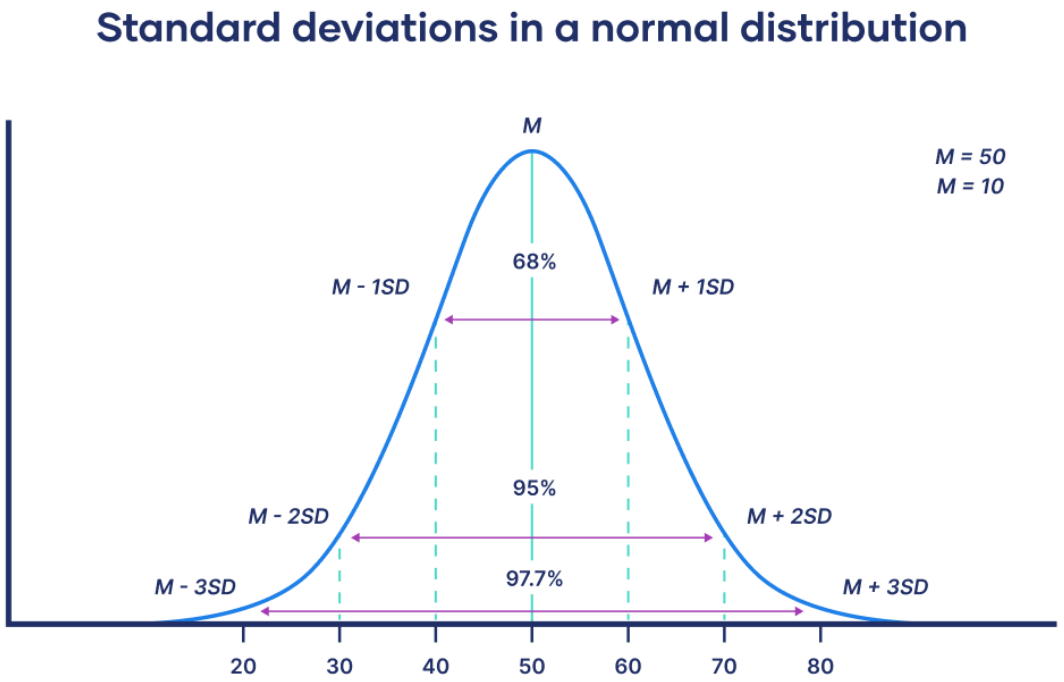
Standard deviation example

- How data is distributed in a **normal distribution**. The empirical rule, or the 68-95-99.7 rule:
- Around 68% of scores are within 1 SD of the mean,
- Around 95% of scores are within 2 SD of the mean,
- Around an.



Standard deviation example

- The data follows a normal distribution with a mean score of 50 and a standard deviation of 10.
- Around 68% of scores are between 40 and 60.
- Around 95% of scores are between 30 and 70.
- Around 99.7% of scores are between 20 and 80.



Mean & standard deviation

- **Chebychev** discovered: with ***any*** list of values, the range mean ± zSD collects ***at least***

$1 - \frac{1}{z^2} \times 100\%$ of values in the list

- Mean ± 1SD collects at least 0% of values
- Mean ± 2SD collects at least 75% of values
- Mean ± 3SD collects at least 88.9% of values

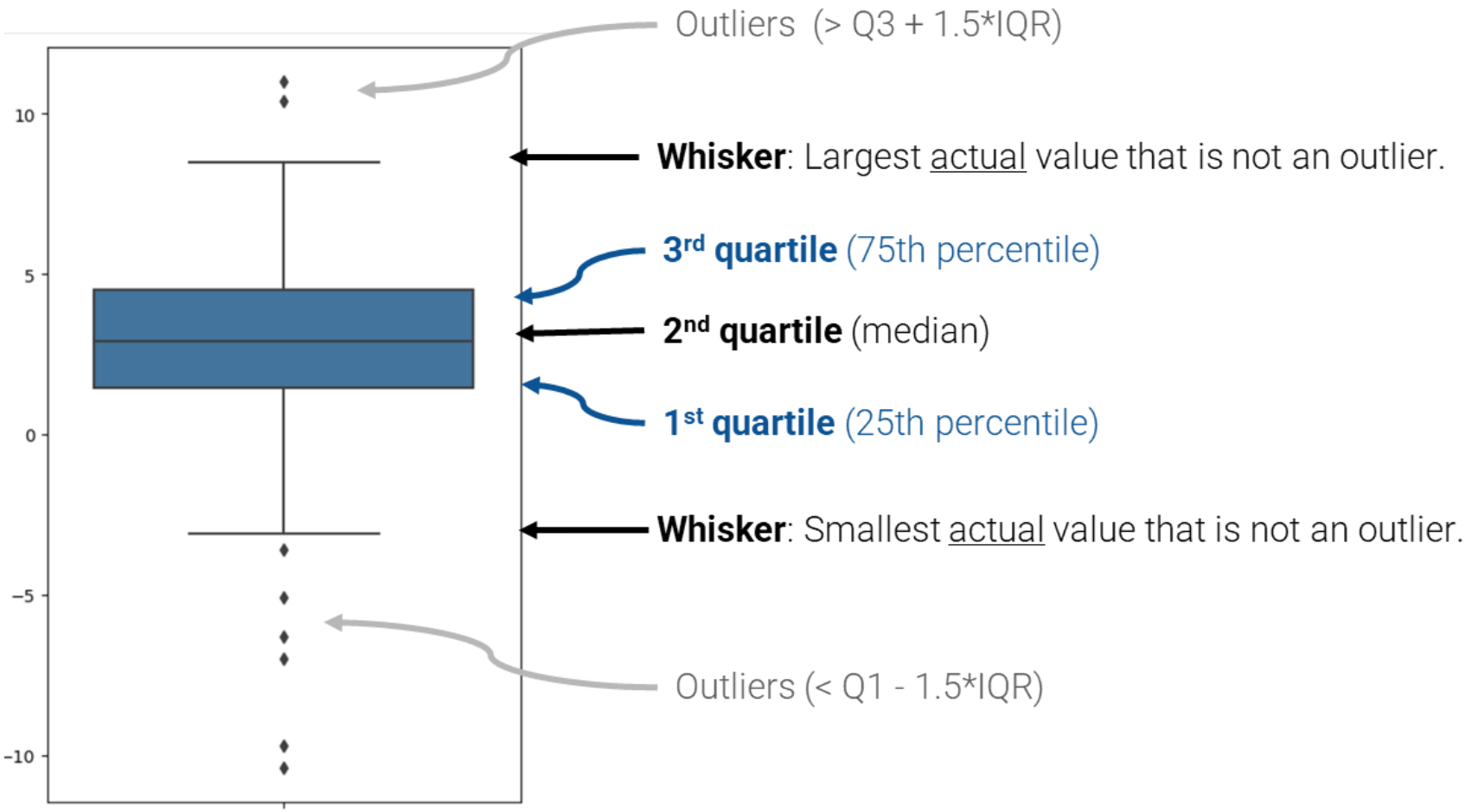
Level 2: quartiles and IQR

	Percentile
Lower quartile Q1	25th
Median Q2	50th
Upper quartile Q3	75th

- **Interquartile range:** IQR = Q3 − Q1, the range of the **middle 50%** of values
- **Five-number summary:** min, Q1, median, Q3, max
- Like the median, the quartiles are **robust**: one extreme value barely moves them
- Rule of thumb for outliers: below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR (Tukey).

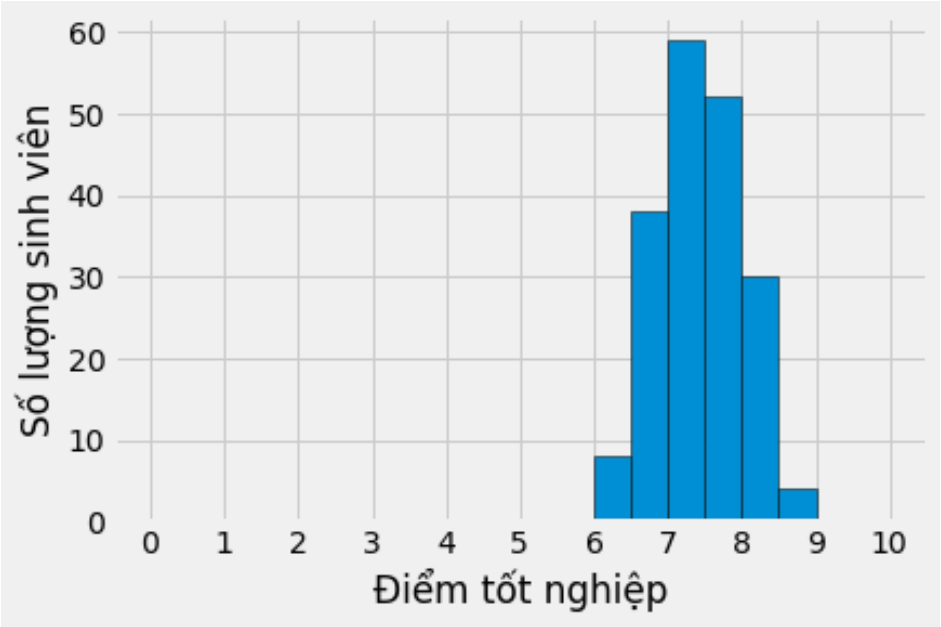
Level 2: quartiles and IQR

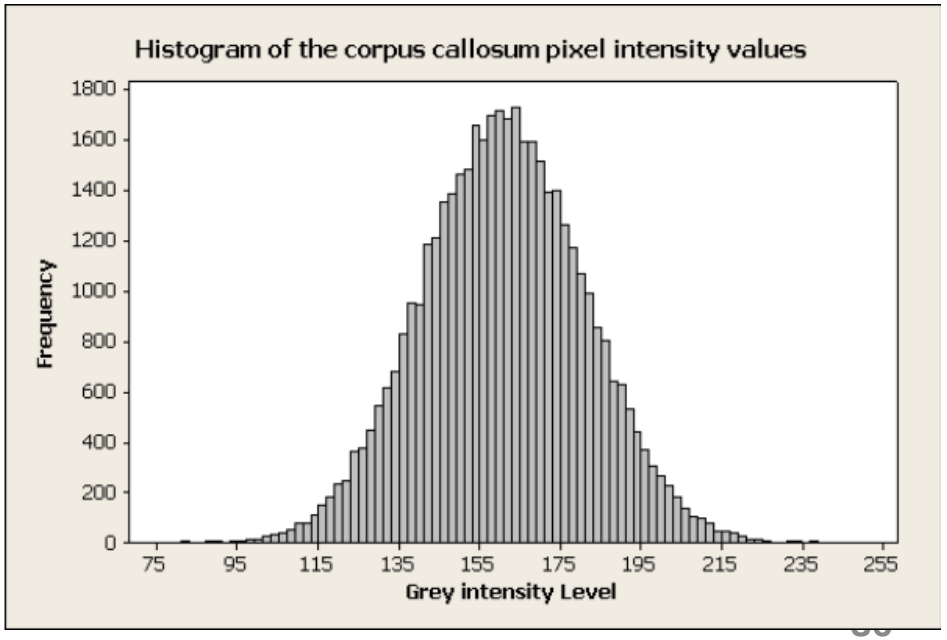
- **Interquartile range:** IQR = Q3 − Q1



Histogram

- **Histogram** is a graphical representation of distribution
- A **frequency distribution** shows how often each different value in a set of data occurs.
- A histogram is the most commonly used graph to show frequency distributions.

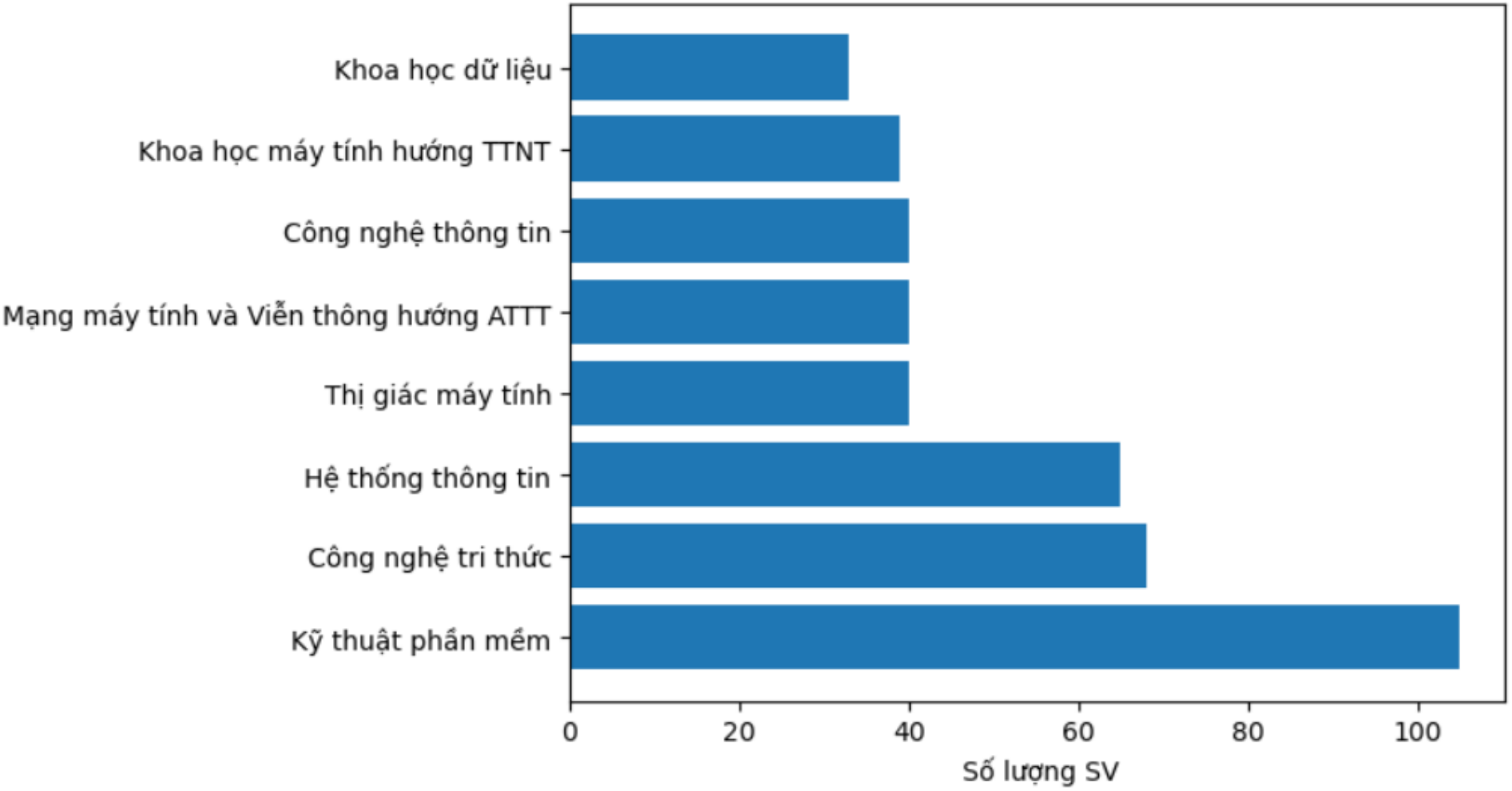




Bar chart

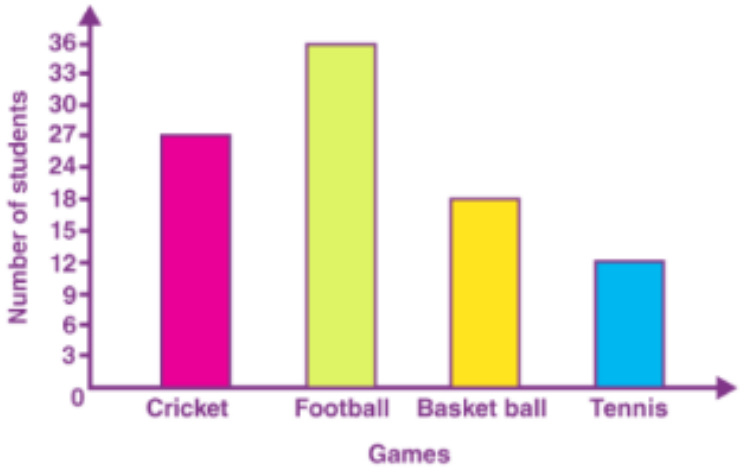
- **Bar chart** is a way of showing the distribution of data with a ***categorical data*** set.
- Bar charts can be represented horizontally or vertically, they can also represent more than one set of data.
- One axis is labelled with the category/group and the other labelled with the frequency of the category/group.

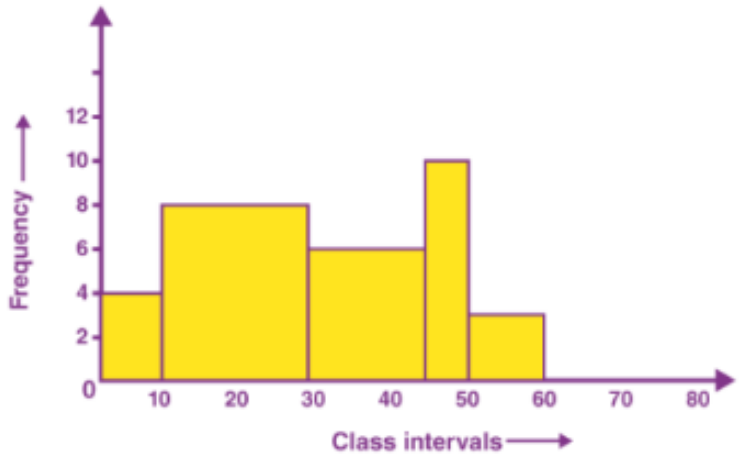
Bar chart


Khoa học dữ liệu
Khoa học máy tính hướng TTNT
Công nghệ thông tin
Mạng máy tính và Viễn thông hướng ATTT
Thị giác máy tính
Hệ thống thông tin
Công nghệ tri thức
Kỹ thuật phần mềm
0
20
40
60
80
100
Số lượng SV

Bar Chart vs. Histogram

Bar chart	Histogram
Graphical representation of **categorical** data	Graphical representation of **numberal** data
There is **equal space** between each pair of consecutive bars	There is **no space** between the consecutive bars
The **height** of the bars shows the frequency, and the width of the bars are **same**	The **area** of rectangular bars shows the frequency and the width of the bars **need not** to be same







THE END